Journal of Alloys and Compounds 1010 (2025) 177343

Contents lists available at ScienceDirect

Journal of Alloys and Compounds

journal homepage: www.elsevier.com/locate/jalcom

Superior piezoelectric performance in ytterbium-substituted high-T_C $Bi_5Ti_3FeO_{15}$

Xin-Yu Yu [a], Xiang-Yi Gao [b], Qian Wang [a], Hui-Lin Li [a], Yi-Jun Wan [a], En-Meng Liang [a], Chun-Ming Wang [a,c,*]

[a] School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan, Shandong 250100, China
[b] Shandong Experimental High School, Jinan, Shandong 250001, China
[c] Center for Optics Research and Engineering (CORE), Key Laboratory of Laser and Infrared System of Ministry of Education, Shandong University, Qingdao, Shandong 266237, China

ARTICLE INFO

Keywords:
Bismuth layer-structured ferroelectrics (BLSFs)
Bismuth titanate-ferrite ($Bi_5Ti_3FeO_{15}$)
High Curie temperature (High-T_C)
Ytterbium oxide (Yb_2O_3)
High-temperature piezoelectric sensors
Piezoelectric ceramics

ABSTRACT

High-temperature piezoelectric materials are essential for advanced sensor applications. However, it is challenging to achieve both high piezoelectric performance and a broad service temperature range. Aurivillius-type bismuth layer-structured ferroelectric (BLSF) bismuth titanate-ferrite ($Bi_5Ti_3FeO_{15}$) has recently received significant attention. However, the piezoelectric properties of $Bi_5Ti_3FeO_{15}$-based compounds have not been extensively explored due to their poor piezoelectric performance and low direct-current (dc) electrical resistivity at elevated temperatures. To address these limitations, we optimized the composition by partially substituting *A*-site bismuth ions with rare-earth ytterbium ions. This substitution aimed to enhance the piezoelectric performance, dc electrical resistivity, and thermal stability of the piezoelectric properties. We conducted a detailed examination of the crystal structure, microstructural morphology, and piezoelectric properties of ytterbium-substituted $Bi_5Ti_3FeO_{15}$. The optimal composition, 4 mol% ytterbium-substituted $Bi_5Ti_3FeO_{15}$ (BTF-4Yb), exhibits a significantly improved piezoelectric constant (d_{33}) of 21.4 pC/N, which is three times higher than that of $Bi_5Ti_3FeO_{15}$ (7.1 pC/N). Additionally, it shows a high dc electrical resistivity of 1.6 MΩ cm at 400 °C and a high Curie temperature (T_C) of 771 °C. Importantly, BTF-4Yb maintains good thermal stability of electromechanical coupling characteristics up to 300 °C. These results indicate that ytterbium-substituted $Bi_5Ti_3FeO_{15}$ is a promising candidate for high-temperature piezoelectric ceramics, offering significant potential for applications in high-temperature piezoelectric sensors.

1. Introduction

Piezoelectric materials, which convert electrical energy into mechanical energy, are highly sought after for piezoelectric applications operating at elevated temperatures (≥ 300 ℃) to ensure structural health monitoring and diagnostics in fields such as automotive, aerospace, and power plants [1]. Currently, lead zirconate titanate (PZT) piezoelectric ceramics with morphotropic phase boundary (MPB) compositions are the predominant choice due to their excellent electromechanical coupling properties. However, PZT ceramics are unsuitable for high-temperature applications due to their relatively low Curie temperature ($T_C < 400$ ℃). Since piezoelectric materials can only be safely used up to approximately half of their T_C without significant performance degradation, PZT ceramics are not viable for applications exceeding 250 ℃ [2]. Aurivillius-type bismuth layer-structured ferroelectrics (BLSFs), such as $CaBi_2Nb_2O_9$ (CBN) [3–6], Bi_3TiNbO_9 (BTN) [7], $CaBi_4Ti_4O_{15}$ (CBT) [8–11], $Bi_4Ti_3O_{12}$ (BIT) [12–16], $Na_{0.5}Bi_{2.5}Nb_2O_9$ (NBN) [17–19], have emerged as promising candidates for high-temperature applications due to their high T_C exceeding 600 ℃. The general formula for BLSF oxides is $(Bi_2O_2)^{2+}(A_{m-1}B_mO_{3m+1})^{2-}$, where *A* represents monovalent, divalent, or trivalent metallic ions suitable for dodecahedral coordination, *B* represents tetravalent, pentavalent, or hexavalent transition metallic ions suitable for octahedral coordination, and *m* ranges from 1 to 5, indicating the number of octahedral layers in the perovskite slab [20–23].

Despite their potential, BLSF compounds typically exhibit weak piezoelectric performance, generally piezoelectric constant (d_{33}) not exceeding 10 pC/N, due to high coercive fields (E_c) and structural

* Corresponding author at: School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan, Shandong 250100, China.
E-mail address: wangcm@sdu.edu.cn (C.-M. Wang).

https://doi.org/10.1016/j.jallcom.2024.177343
Received 8 August 2024; Received in revised form 1 October 2024; Accepted 30 October 2024
Available online 31 October 2024
0925-8388/

anisotropy [24]. Additionally, BLSFs often suffer from low electrical resistivity, attributed to high concentrations of oxygen vacancies resulting from the volatilization of bismuth during sintering [25,26]. To address these limitations, researchers have explored various methods to enhance the piezoelectric properties of BLSF compounds, such as chemical doping and grain orientation techniques [27–29].

Bismuth titanate-ferrite ($Bi_5Ti_3FeO_{15}$, BTF) with $m = 4$ stands out in the BLSF family as a promising high-temperature piezoelectric material, due to its high T_C (~761 °C) [30]. While recent studies have focused on the electromagnetic and ferromagnetic coupling applications of BTF, there has been limited research on its piezoelectric properties. This is largely attributed to the material's rigid poling conditions, such as high E_c and significant leakage currents. Nevertheless, chemical modification has proven to be a cost-effective strategy for simultaneously enhancing the piezoelectric properties and electrical resistivity of BTF, paving the way for large-scale manufacturing. For example, Moure et al. reported an enhanced d_{33} of 18.6 pC/N in $Bi_5Ti_3FeO_{15}$-$CaBi_4Ti_4O_{15}$ (BTF-CBT) [31]; Huang et al. achieved a d_{33} of 13.4 pC/N in (Na, Ce) co-modified BTF [32]; and Wang et al. attained a high d_{33} of 23 pC/N in Mn-modified BTF [30].

However, there is a paucity of studies on the effects of *A*-site rare-earth ion substitutions on the piezoelectric properties and thermal stability of BTF. In this study, we propose a strategy involving *A*-site substitution with ytterbium (Yb) to stabilize the $(Bi_2O_2)^{2+}$ layer and reduce the concentration of oxygen vacancies. This approach aims to enhance both the piezoelectric properties and thermal stability of BTF. We thoroughly investigate the structure, dielectric, electrical, piezoelectric, and electromechanical properties of *A*-site ytterbium-substituted BTF.

2. Experimental procedure

Ytterbium-substituted $Bi_5Ti_3FeO_{15}$ with nominal compositions of $Bi_{5\text{-}x}Yb_xTi_3FeO_{15}$ (abbreviated as BTF-100*x*Yb, where $x = 0.00$, 0.02, 0.04, 0.06, 0.08, 0.10, 0.15, and 0.20) were fabricated using the conventional solid-state reaction method. Analytical grade Bi_2O_3 (99.80 %), TiO_2 (99.80 %), Fe_2O_3 (99.40 %), and Yb_2O_3 (99.99 %) were used as raw materials and weighed in stoichiometric ratios. The raw materials were mixed and milled with ZrO_2 balls in ethanol for 12 h. After drying, the mixture was calcined at 750 °C for 3 h to synthesize the target product. The calcined mixture was then ground, milled, and dried again under the same conditions. The resulting powder was granulated with polyvinyl alcohol (PVA) binders and pressed into disks 12 mm in diameter and 1.0 mm in thickness. Finally, each disk was heated at 650 °C to burn off the PVA binder and then sintered at 1050–1060 °C for 3 h at a heating rate of 3 °C/min in an aluminum oxide crucible to prepare the ytterbium-substituted BTF ceramic samples.

The structural analysis of the prepared powders was conducted using an X-ray diffractometer (XRD, Smartlab SE; Rigaku Corporation) with Cu Kα radiation. Raman spectra of the samples were obtained using a Raman spectrometer (LabRAM HR 800, HORIBA Jobin-Yvon) with a He-Ne laser (632.8 nm). Natural surface micrographs were captured using a field-emission scanning electron microscope (FE-SEM, Merlin FE-SEM; Carl Zeiss). Prior to measuring electrical properties, silver electrodes were screen-printed on both surfaces of polished samples and fired at 570 °C for 20 min. Dielectric properties and resonance response were measured using an impedance analyzer (E4990A; Keysight Technologies) equipped with a multimeter (Keithley 2000; Tektronix). Electrical resistivity was calculated from resistance measurements taken by a source meter (Keithley 2410; Tektronix Inc.) with a voltage of 100 V applied to the sample. Ferroelectric hysteresis loops were obtained using a TF 2000E Analyzer (aixACCT Systems GmbH, Germany). The piezoelectric constant d_{33} was measured using a YE2730A d_{33} meter after the samples were poled in silicone oil under a direct-current (dc) electric field of 60–70 kV/cm at 150 °C for 20 min.

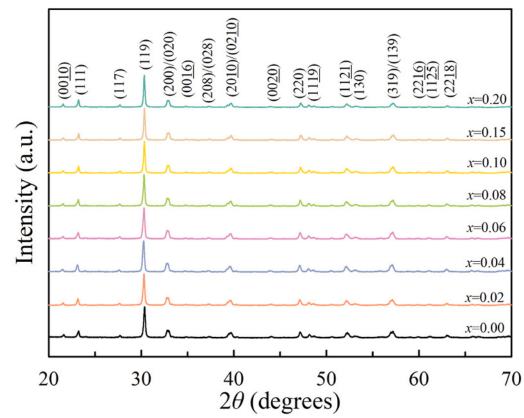

Fig. 1. X-ray diffraction patterns of BTF-100*x*Yb ceramics.

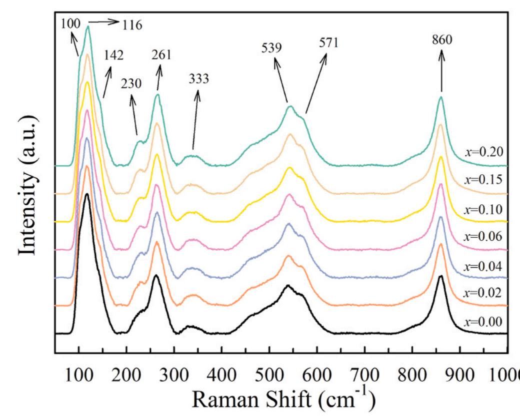

Fig. 2. Raman spectra of unmodified and ytterbium-substituted BTF ceramics.

3. Results and discussion

Fig. 1 shows the X-ray diffraction patterns of the BTF-100*x*Yb ceramics. All diffraction peaks are well indexed by the standard diffraction data [33], illustrating a single orthorhombic phase with the space group $A2_1am$. No impurity phases were detected in any of the samples, implying that the ytterbium dopant ions were successfully incorporated into the pseudo-perovskite layers, presumably occupying the Bi^{3+} at *A*-site. According to Shannon's law [34], due to the difference in effective ionic radius of Yb^{3+} (115 pm, coordination number CN = 12) and Bi^{3+} (136 pm, CN = 12), the substitution of ytterbium results in a lattice distortion, which is responsible for the intrinsic contribution to the significant enhancement in piezoelectric properties of BTF. The strongest diffraction peak (119) of BTF is consistent with the crystal surface index $(11\underline{2m+1})$, corresponding to the strongest diffraction peak of BLSFs [35].

Compared with X-ray diffraction (XRD), Raman scattering is not only more sensitive and efficient for investigating lattice vibration modes but also provides more detailed information related to the crystallographic positions of substitutional ions and their effects on ferroelectric behavior through the occurrence of extra modes and/or shifts in peak position [36]. Fig. 2 presents Raman scattering spectra with Raman shifts ranging from 50 to 1000 cm^{-1} at room temperature. Generally, the Raman spectra of Aurivillius-type materials can be categorized into two types of phonon modes [37]. The first type includes modes with frequencies below 200 cm^{-1}, which are associated with the vibrations of Bi atoms at the *A*-site. The second type includes modes with frequencies above 200 cm^{-1}, related to the internal vibrations of oxygen octahedra in the perovskite-like layers. In the BTF ceramic samples, prominent Raman modes appear at around 100, 116, 142, 230, 261, 333, 539, 571, and

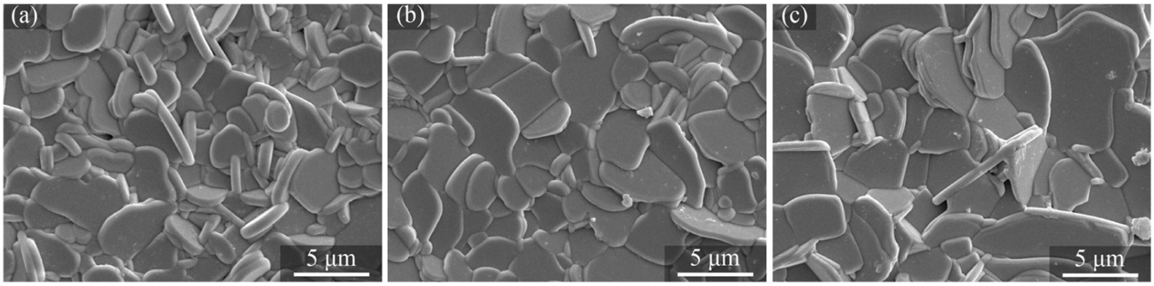

Fig. 3. Scanning electron microscopy images of (a) BTF, (b) BTF-4Yb, and (c) BTF-20Yb.

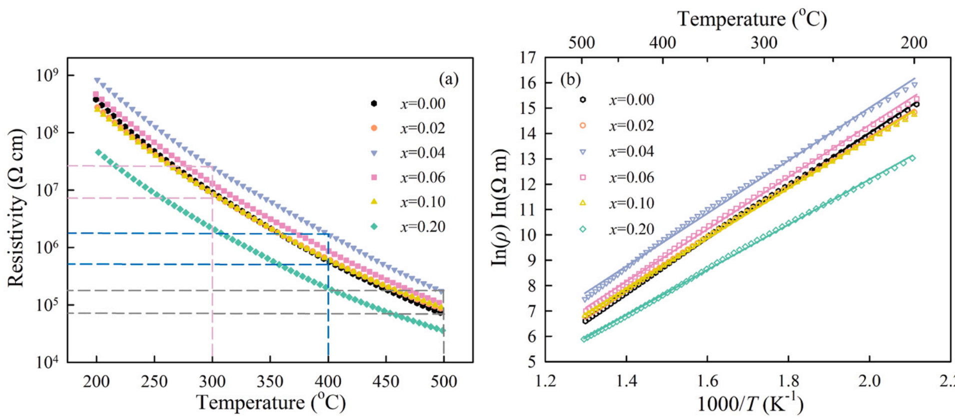

Fig. 4. (a) Temperature dependence of dc electrical resistivity for the unmodified and ytterbium-substituted BTF and (b) the Arrhenius relationships between dc electrical resistivity and temperature.

860 cm^{-1}, indicating the formation of a well-defined four-layer perovskite structure. The low-frequency Raman modes near 100, 116, and 142 cm^{-1} are closely related to the vibrations of Bi^{3+} ions at the *A*-site of the perovskite-like slabs [38]. The high-frequency Raman modes around 230 and 261 cm^{-1} correspond to the bending vibrations of the O-Ti/Fe-O bonds in the oxygen octahedron [39]. Raman modes near 333 cm^{-1} are associated with the combined twisting and stretching vibrations of the oxygen octahedron [40], while the modes around 539 and 571 cm^{-1} are related to the relative vibrations of the oxygen atom at the top outer end of the $(Ti/Fe)O_6$ octahedron [41]. The high-frequency mode appearing around 860 cm^{-1} corresponds to the tensile vibrations of the TiO_6 octahedron along the *c*-axis [42].

After the introduction of ytterbium ions, the characteristic peaks mentioned above remain evident, suggesting that the ytterbium substitution does not significantly alter the basic structure of BTF. Notably, the high-frequency Raman modes located near 230 and 333 cm^{-1} are intensified compared to the BTF, indicating the effect of ytterbium substitution on the distortion of the $(Ti/Fe)O_6$ octahedron. This also suggests that ytterbium ions tend to replace bismuth ions, affecting the local structural environment around the BO_6 octahedron, including changes in the phonon vibration frequency of the outer oxygen atom at the top of the $(Ti/Fe)O_6$ octahedron and a reduction in local structural symmetry. Additionally, the positions and intensities of other Raman peaks show little change, indicating they are not significantly affected by the ytterbium ion replacement of bismuth ions.

Fig. 3 shows scanning electron microscopy (SEM) images of natural surfaces for BTF-100*x*Yb ceramics. All ceramic samples exhibit a compact, pore-free microstructure, indicating optimal sintering conditions for BTF-100*x*Yb ceramics. The grains in all samples are well-developed, laminar, and randomly oriented, characteristic of BLSF ceramics [43]. Due to the high grain growth rate perpendicular to the *c*-axis of BLSF crystals, the grain growth is structurally anisotropic, with the length of the plate-like grains significantly exceeding their thickness. Compared to BTF, the ytterbium-substituted BTF ceramics exhibit larger grain sizes, which are advantageous for enhancing piezoelectric properties [44]. The average grain size of BTF increases significantly with increasing ytterbium content, demonstrating that the substitution of ytterbium improves mass diffusion and accelerates the migration of ions during the sintering process [21], thus facilitating the grain growth. This observation is consistent with recent reports on other BLSF compounds [30,45].

High dc electrical resistivity at elevated temperatures is crucial for the practical application of piezoelectric ceramics. Ceramics with low electrical resistivity can generate high leakage currents, leading to charge drift and interference with the detection of charges induced by the piezoelectric effect [46]. Fig. 4(a) shows the temperature dependence of the dc electrical resistivity of BTF-100*x*Yb ceramics. The dc resistivity gradually decreases with increasing temperature, indicating a thermally activated process with a negative temperature coefficient of dc electrical resistance. As shown in Table 2, the maximum resistivity values are observed for BTF-4Yb, measuring 2.4×10^7 Ω cm at 300 °C, 1.6×10^6 Ω cm at 400 °C, and 1.8×10^5 Ω cm at 500 °C. These values are higher than those of BTF, which are 8.8×10^6 Ω cm at 300 °C, 5.6×10^5 Ω cm at 400 °C, and 6.7×10^4 Ω cm at 500 °C. The high dc electrical resistivity allows ytterbium-substituted BTF ceramics to withstand higher electric fields, enhancing their piezoelectric properties through sufficient polarization. Several factors may contribute to the change in dc resistivity with different levels of ytterbium substitution. The $(Bi_2O_2)^{2+}$ layers play a crucial role in space-charge compensation in Aurivillius-type compounds, improving electrical properties by suppressing leakage currents [47]. Bismuth ions tend to volatilize during high-temperature sintering due to its relatively low melting point (~820 °C), leading to the formation of oxygen vacancies that act as charge carriers [48]. The substitution of the highly volatile bismuth ions with ytterbium ions, which have a higher melting point (~2346 °C), can effectively suppress bismuth loss during sintering. This stabilization of

Table 1
The dc resistivity at high temperature and the activation energy (E_a) of the BTF-100xYb ceramics.

Compositions	ρ (Ω cm)			E_a (eV)
	300 °C	400 °C	500 °C	
BTF	8.8×10^{6}	5.6×10^{5}	6.7×10^{4}	0.90
BTF−2Yb	8.3×10^{6}	6.0×10^{5}	8.7×10^{4}	0.86
BTF−4Yb	2.4×10^{7}	1.6×10^{6}	1.8×10^{5}	0.90
BTF−6Yb	1.3×10^{7}	8.7×10^{5}	1.1×10^{5}	0.89
BTF−10Yb	8.3×10^{6}	5.8×10^{5}	9.1×10^{4}	0.85
BTF−20Yb	2.0×10^{6}	1.9×10^{5}	3.6×10^{4}	0.77

the $(Bi_2O_2)^{2+}$ layers reduced the formation of point defects such as oxygen vacancies. Another contributing factor is the change in grain morphology due to ytterbium modifications. In Aurivillius-type materials, grain growth is structurally anisotropic, occurring primarily in the *a-b* plane, while the $(Bi_2O_2)^{2+}$ layers along the *c*-axis act as natural insulating media. Larger grain sizes in ytterbium-substituted ceramics facilitate long-range charge transport due to reduced grain boundary effects and weaker insulation by the $(Bi_2O_2)^{2+}$ layers [8]. This explains why the resistivity decreases with increasing ytterbium substitution, particularly in BTF-20Yb.

The temperature-dependent resistivity behavior follows the Arrhenius relationship [49]: $\rho = \rho_0 \exp(E_a/k_BT)$, where ρ_0 is a pre-exponential factor, E_a is the activation energy, k_B is the Boltzmann constant, and T is the absolute temperature. The fitted results are shown in Fig. 4(b) and Table 1. The activation energy E_a, calculated from the fitting slopes, first increases and then decreases with increasing ytterbium substitution, mirroring the trend in dc resistivity. An activation energy of ~0.90 eV, typical for the migration enthalpy of oxygen, suggests that the electrical conduction process in BTF-100xYb ceramics is dominated by the oxygen vacancies [50].

Fig. 5 shows the dielectric constant (ε_r) and dielectric loss (tanδ) of BTF-100xYb measured at 1 MHz. The ε_r of all samples increases gradually as the temperature rises from room temperature (RT) to 500 °C. However, it increases sharply above 500 °C due to high electrical conductivity [24]. All samples exhibit a distinct peak in ε_r, corresponding to the ferroelectric-to-paraelectric phase transition, i.e., the T_C of the material. Although only one phase transition is observed in BTF-100xYb, BTF-20Yb shows a small bump around 600 °C, attributed to space charge effects, which is consistent with observations in Co-doped BTF [51]. As indicated in the inset of Fig. 5(a) and Table 2, T_C values significantly increase with ytterbium substitution, due to the distortion of the BTF lattice caused by the introduction of ytterbium ions. The optimal composition, BTF-4Yb, has a T_C of 771 °C, and the maximum T_C of 805 °C is achieved for BTF-20Yb, which is 42 °C higher than that of unmodified BTF (763 °C). Previous studies have shown that T_C values in BLSFs are strongly related to the degree of structural distortion [52–55], described by the tolerance factor (t), defined as $t = \frac{R_A+R_O}{\sqrt{2}(R_B+R_O)}$, where R_A, R_B, and R_O are the ionic radii of A-site cations, B-site cations, and oxygen ions, respectively. Typically, the tolerance factor t is inversely proportional to T_C [56]. According to Shannon's law [34], compared with Bi^{3+} (136 pm, CN = 12) at the A-site, the substitution of Yb^{3+} (115 pm, CN = 12) with a smaller ionic radius at the A-site leads to a decrease in t and an increase in T_C. Similarly, as shown in Fig. 5(b), the tanδ increases gradually up to 500 °C but rises sharply above 500 °C due to high conductivity and thermally activated domain wall movements [3]. Research suggests that oxygen vacancies ($V_O^{\cdot\cdot}$) resulting from bismuth volatilization during the calcination process act as space charges, leading to the domain pinning effect and increased tanδ [57]. The substitution of ytterbium reduces the oxygen vacancy concentration, improves the dc resistivity, and thus reduces the conductivity loss [55]. In addition, the substitution of ytterbium can cause charges to accumulate in specific regions, disrupting the long-range ordering in the ceramics [58] and thereby increasing the contribution of defect-induced losses. With the combination of these two mechanisms, the dielectric loss increased with more ytterbium substitutions as the temperature increases. However, the dielectric loss of BTF-4Yb ceramics is lower than 0.03 below 300 °C, which is conducive to its application in high temperature scenarios.

Benefitting from the slight lattice distortion, increased grain size, and reduced oxygen vacancy concentration induced by the introduction of ytterbium, the piezoelectric and electromechanical properties of BTF-100xYb ceramics are significantly improved. As listed in Table 2, the piezoelectric constant d_{33} is prominently improved and shows a trend of first increasing and then decreasing with the increasing ytterbium substitution. The optimal piezoelectric performance with a maximum d_{33} value of 21.4 pC/N is achieved by BTF-4Yb, which is more than three times that of the unmodified BTF of 7.1 pC/N. In addition, an enhanced planar frequency constant (N_p) of 2096 Hz m, the optimal electromechanical coupling factor (k_p) of 5.1 % and a mechanical quality factor (Q_{mp}) of 2550 are obtained simultaneously in BTF-4Yb. Table 3 lists the

Table 2
Dielectric and piezoelectric properties of the BTF-100xYb ceramics.

Compositions	T_C (°C)	ε_r	tanδ (%)	d_{33} (pC/N)	k_p (%)	N_p (Hz m)	Q_{mp}
BTF	763	115	0.45	7.1	3.2	2073	2460
BTF−2Yb	768	129	0.42	20.1	4.5	2086	2110
BTF−4Yb	771	127	0.34	21.4	5.1	2096	2550
BTF−6Yb	780	127	0.38	20.4	4.5	2119	2190
BTF−8Yb	784	121	0.32	19.7	4.5	2118	2110
BTF−10Yb	788	123	0.42	21.0	5.0	2089	1990
BTF−15Yb	800	128	0.35	17.0	4.7	2104	1830
BTF−20Yb	805	126	0.42	14.0	4.5	2113	1860

*The Data were measured at room temperature at 1 MHz.

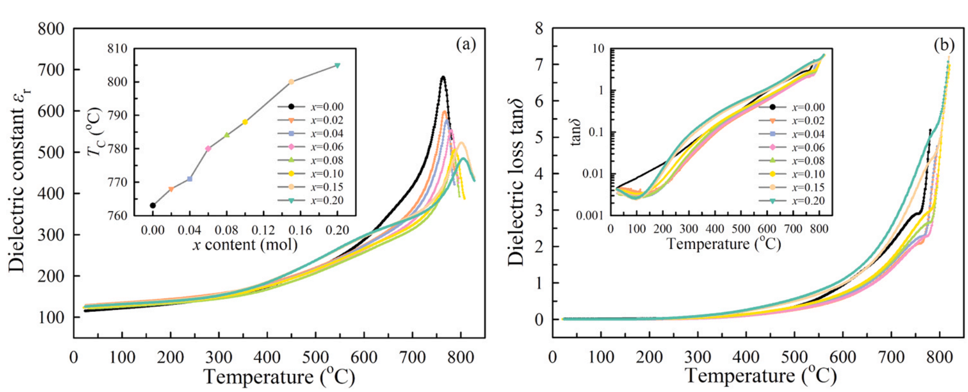

Fig. 5. Temperature-dependent (a) dielectric constant (ε_r) and (b) dielectric loss (tanδ) of BTF-100xYb ceramics. The inset of (a) shows the relationship between T_C and x content. The inset of (b) shows the logarithmic form of the tanδ change with temperature.

Table 3
Comparison of T_C and d_{33} of recently reported BTF-based ceramics.

Compositions	T_C (℃)	d_{33} (pC/N)	Ref.
$Bi_{4.96}Yb_{0.04}Ti_3FeO_{15}$	771	21.4	This work
$Bi_5Ti_3Fe_{0.96}Mn_{0.04}O_{15}$	765	23.0	[30]
$0.75Bi_5Ti_3FeO_{15}-0.25CaBi_4Ti_4O_{15}$	730	18.6	[31]
$Bi_{4.8}(NaCe)_{0.1}Ti_3FeO_{15}$	743	13.4	[32]
$Bi_5Ti_{2.2}Fe_{1.4}Nb_{0.4}O_{15}$	-	10.0	[59]

comparison of d_{33} and T_C of the reported BTF-based ceramics. In this work, a high d_{33} of 21.4 pC/N with a large T_C of 771 ℃ is achieved in BTF-4Yb, which is superior to that of previous works [30–32,59]. Based on these properties, it can be shown that ytterbium-substituted BTF ceramics are excellent candidates for high-temperature applications.

To investigate the macroscopic ferroelectric behaviors of BTF-100xYb ceramics, Fig. 6(a) and (b) present the polarization-electric field (*P-E*) loops and electric current density-electric field (*J-E*) curves, measured at 160 kV/cm, 1 Hz, and 150 °C. All BTF-100xYb ceramic samples exhibit typical ferroelectric *P-E* loops, indicating the formation of good ferroelectric properties. Compared to unmodified BTF, the *P-E* loop for BTF-4Yb shows more saturated states, while the *P-E* loops become slimmer and spindly as ytterbium content increases to 20 mol%. As shown in Fig. 6(b), the current peaks in the *J-E* loop correspond to the coercive field (E_C) due to the reversal of spontaneous dipoles. Compared to the unmodified BTF, the current peaks of ytterbium-substituted ceramics shift to higher electric fields. As listed in Table 4, the remanent polarization (P_r) of the BTF-100xYb ceramics increases as x rises from 0 to 0.04, before decreasing as x increases further to 0.20, reaching its maximum for BTF-4Yb. Generally, enhanced piezoelectric performance is often associated with increased spontaneous or remnant polarization (P_s or P_r) [60]. The largest remnant polarization of BTF-4Yb suggests excellent piezoelectricity. For ytterbium-substituted BTF ceramics, the improvement in piezoelectric properties compared to unmodified BTF seems to be primarily due to slight lattice distortion, increased grain size, improved insulation, and reduced oxygen vacancy concentration. Additionally, it is worth noting that the symmetry of the *P-E* loop is not ideal ($E_{c+} \neq E_{c-}$), which can be attributed to the presence of an internal bias field induced by an increase in the number of defective dipole pairs within the material [61].

Fig. 7 illustrates the thermal annealing behavior of BTF-100xYb ceramics to evaluate their piezoelectric performance after annealing treatment. In the low-annealing temperature range (< 300 °C), the d_{33} values show a slight decrease but retain more than 90 % of their original values due to thermally activated reverse switching in the non-180 ° domains [62]. As the temperature increases from 300 to 600 °C, the d_{33} values decrease more rapidly with higher ytterbium doping but still maintain above 80 % of their original values. Oxygen vacancies play a role in pinning the domains, making domain reversal difficult [63]. However, the introduction of ytterbium stabilizes the $(Bi_2O_2)^{2+}$ layer and reduces the oxygen vacancy content, leading to a quicker decrease in d_{33} values. Notably, the ytterbium-substituted ceramics retain a high piezoelectric constant (d_{33} > 75 % of the original values) even after annealing at 700 °C, whereas the unmodified BTF almost completely loses its piezoelectricity. This result demonstrates the potential of ytterbium-substituted ceramics for applications in high-temperature environments.

Table 4
The ferroelectric parameters P_{max}, P_r and E_c under a driving electric field of 160 kV/cm for unmodified and ytterbium-substituted BTF ceramics.

Compositions	P_{max} (μC/cm^2)	P_r (μC/cm^2)	E_{c+} (kV/cm)	E_{c-} (kV/cm)
BTF	8.89	3.48	46.84	57.53
BTF−2Yb	10.89	5.94	64.70	74.40
BTF−4Yb	12.84	6.18	50.20	65.98
BTF−8Yb	7.63	3.62	54.42	68.04
BTF−10Yb	4.92	3.13	75.44	85.58
BTF−20Yb	5.49	2.56	62.92	70.49

To study the thermal stability of the piezoelectric performance in ytterbium-substituted BTF, we examined the temperature-dependent electrical impedance of BTF-4Yb, as shown in Fig. 8, where the frequency-dependent electromechanical resonance characteristics (k_p mode) at elevated temperatures are shown. At room temperature, the resonance and antiresonance frequencies appear near 252 kHz but shift steadily toward lower frequencies with decreasing impedance amplitude as the temperature increases from 30 °C to 500 °C. At lower temperatures (T < 300 °C), the electrical impedances of the resonance peaks and valleys are Z(f_a) ~ 22.2 kΩ and Z(f_r) ~ 2.4 kΩ at RT, and Z(f_a) ~ 5.4 kΩ and Z(f_r) ~ 3.9 kΩ at 300 °C, respectively, showing a considerable contrast between Z(f_a) and Z(f_r). However, above 300 °C, the resonance amplitude decreases sharply due to increased electrical conductivity [14]. Despite the weak resonance/antiresonance amplitude at 500 °C, the resonance characteristics are still evident, indicating that BTF-4Yb retains its piezoelectric nature up to 500 °C. Based on the temperature-dependent impedance spectra, the measured resonance frequency (f_r) and antiresonance frequency (f_a) as a function of temperature are shown in Fig. 8(c) and (d). From RT to 300 °C, both f_r and f_a shift almost linearly to lower frequencies with increasing temperature. However, above 300 °C, the resonance frequency changes dramatically due to increased conductivity. The temperature coefficients of the resonance and antiresonance frequencies are found to be approximately 90 ppm/K. Despite high temperatures, BTF-4Yb retains its resonance characteristics. It is important to note that high conductivity in a lossy resonator invalidates the equivalent electrical circuit of a piezoelectric vibrator [64], making most equations inapplicable for calculating piezoelectric parameters. Nonetheless, BTF-4Yb demonstrates good thermal stability of piezoelectric properties, indicating its potential for high-temperature piezoelectric applications.

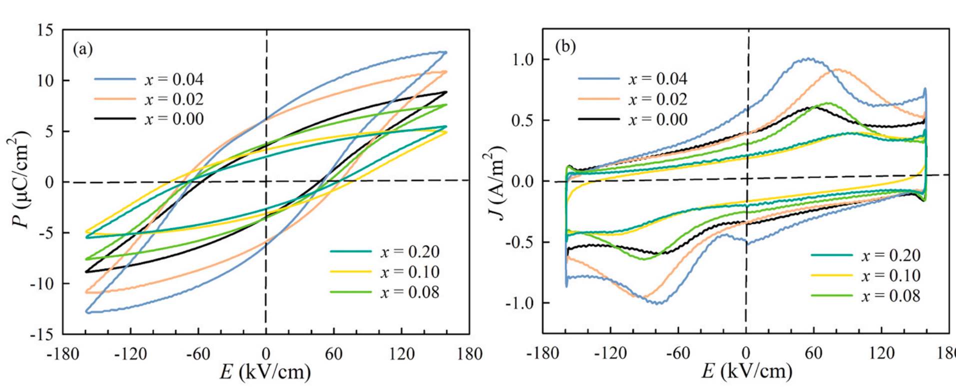

Fig. 6. (a) Polarization-electric field (*P-E*) loops and (b) current density-electric field (*J-E*) loops for unmodified and ytterbium-substituted BTF ceramics.

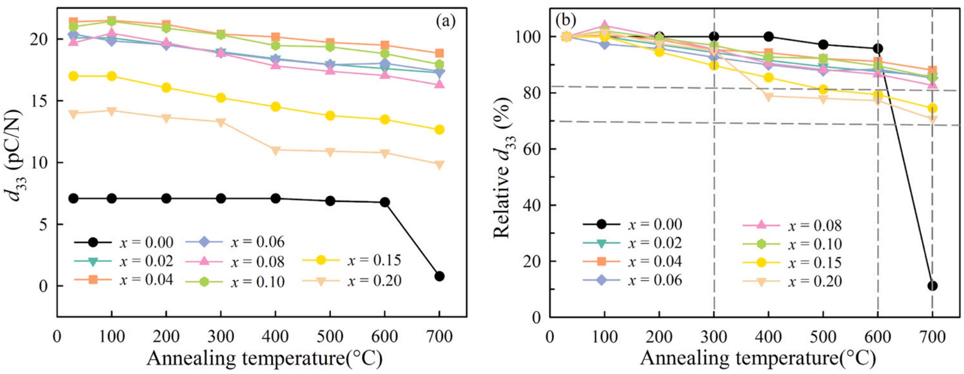

Fig. 7. (a) Thermal annealing behavior of piezoelectric performance and (b) the corresponding relative d_{33} values of BTF-100xYb ceramics.

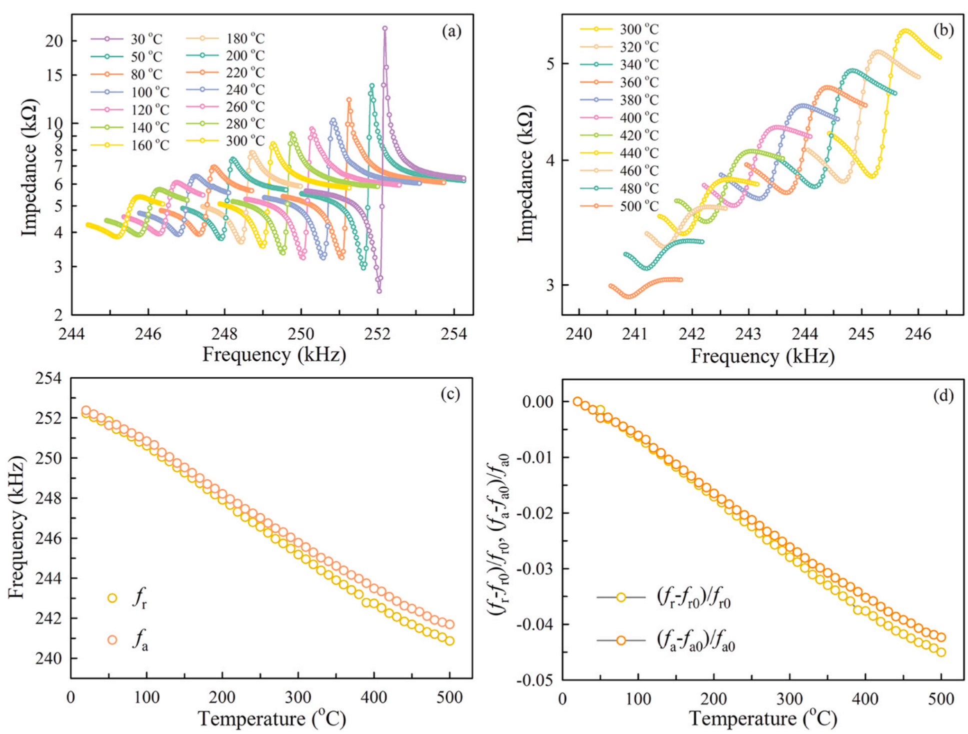

Fig. 8. Temperature-dependent impedance |Z| of BTF-4Yb ceramics as *a* function of frequency: (a) 30 °C to 300 °C, and (b) 300 °C to 500 °C. Temperature-dependent resonance frequency f_r and anti-resonance frequency f_a of BTF-4Yb in (c) and (d).

4. Conclusion

Bismuth layer-structured ferroelectric $Bi_5Ti_3FeO_{15}$ ceramics with rare-earth ytterbium ion substitutions were synthesized using the conventional solid-state reaction method. We thoroughly investigated the structural, dielectric, electrical, piezoelectric, and electromechanical properties of *A*-site ytterbium-substituted $Bi_5Ti_3FeO_{15}$. The piezoelectric constant (d_{33}) was significantly enhanced by substituting ytterbium ions for bismuth ions, with BTF-4Yb exhibiting the optimal piezoelectric performance ($d_{33} = 21.4$ pC/N), which is three times greater than that of BTF. The Curie temperature (T_C) values slightly increased with higher ytterbium content due to the lattice distortion of the BTF caused by substituting smaller radius ytterbium ions for bismuth ions. Temperature-dependent dc electrical resistivity measurements demonstrated that proper ytterbium introduction can improve dc electrical resistivity, with BTF-4Yb showing the highest dc electrical resistivity among all the compositions. The thermal annealing behavior of BTF-100xYb ceramics indicates that ytterbium-substituted BTF ceramics have potential for high-temperature applications. Additionally, the temperature-dependent electrical impedance, resonance frequency, and anti-resonance frequency measurements show that BTF-4Yb maintains good thermal stability of electromechanical properties. These findings suggest that ytterbium-substituted BTF ceramics are promising materials for high-temperature piezoelectric applications.

CRediT authorship contribution statement

Xin-Yu Yu: Writing – original draft, Investigation, Formal analysis, Data curation. **Qian Wang:** Writing – original draft, Investigation, Formal analysis. **Xiang-Yi Gao:** Writing – original draft, Investigation, Data curation. **Chun-Ming Wang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Hui-Lin Li:** Investigation, Data curation. **Yi-Jun Wan:** Investigation, Data curation. **En-Meng Liang:** Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Number: 52372115), Opening Project of Key Laboratory of Inorganic Functional Materials and Devices, Chinese Academy of Sciences (Grant Number: KLIFMD202309), and the Fundamental Research Funds for the Central Universities.

Data Availability

Data will be made available on request.

References

[1] S. Zhang, F. Yu, Piezoelectric materials for high temperature sensors, J. Am. Ceram. Soc. 94 (2011) 3153–3170.
[2] C.A. Randall, N. Kim, J.-P. Kucera, W. Cao, T.R. Shrout, Intrinsic and extrinsic size effects in fine-grained morphotropic-phase-boundary lead zirconate titanate ceramics, J. Am. Ceram. Soc. 81 (1998) 677–688.
[3] J.-N. Chen, Q. Wang, X. Zhao, C.-M. Wang, Significantly enhanced dc electrical resistivity and piezoelectric properties of Tb-modified $CaBi_2Nb_2O_9$ ceramics for high-temperature piezoelectric applications, J. Am. Ceram. Soc. 105 (2022) 4815–4826.
[4] C.-M. Wang, J.-F. Wang, S. Zhang, T.R. Shrout, Piezoelectric and electromechanical properties of ultrahigh temperature $CaBi_2Nb_2O_9$ ceramics, Phys. Status Solidi (RRL) - Rapid Res. Lett. 3 (2009) 49–51.
[5] C.-M. Wang, S. Zhang, J.-F. Wang, M.-L. Zhao, C.-L. Wang, Electromechanical properties of calcium bismuth niobate ($CaBi_2Nb_2O_9$) ceramics at elevated temperature, Mater. Chem. Phys. 118 (2009) 21–24.
[6] C.-M. Wang, J.-N. Chen, H.-T. Lu, Q. Wang, High Curie temperature bismuth layer-structured ferroelectric calcium bismuth niobate ($CaBi_2Nb_2O_9$) piezoelectric ceramics, Chin. Sci. Bull. 66 (2021) 2061–2070.
[7] Q. Wang, C.-M. Wang, J.-F. Wang, S. Zhang, High performance Aurivillius-type bismuth titanate niobate (Bi_3TiNbO_9) piezoelectric ceramics for high temperature applications, Ceram. Int. 42 (2016) 6993–7000.
[8] Z.-Y. Shen, W.-Q. Luo, Y. Tang, S. Zhang, Y. Li, Microstructure and electrical properties of Nb and Mn co-doped $CaBi_4Ti_4O_{15}$ high temperature piezoceramics obtained by two-step sintering, Ceram. Int. 42 (2016) 7868–7872.
[9] G. Li, L. Zheng, Q. Yin, B. Jiang, W. Cao, Microstructure and ferroelectric properties of MnO_2-doped bismuth-layer $(Ca,Sr)Bi_4Ti_4O_{15}$ ceramics, J. Appl. Phys. 98 (2005) 064108.
[10] J. Zeng, Y. Wang, Y. Li, Q. Yang, Q. Yin, Ferroelectric and piezoelectric properties of tungsten doped $CaBi_4Ti_4O_{15}$ ceramics, J. Electroceram. 21 (2008) 305–308.
[11] M.S. Islam, J. Kano, S. Tsukada, Q. Yin, S. Kojima, Microstructure and Phase Transition of MnO_2-Doped Bismuth Layered-Structure Ferroelectrics, Jpn. J. Appl. Phys. 48 (2009), 09KC10.
[12] F. Zhang, W. Shi, S. Guan, Y. Xu, H. Yang, H. Chen, J. Xing, H. Liu, Q. Chen, Enhanced electrical properties and thermal stability of W/Cr co-doped BIT-based high-temperature piezoelectric ceramics, J. Alloy. Compd. 907 (2022) 164492.
[13] H.S. Shulman, M. Testorf, D. Damjanovic, N. Setter, Microstructure, Electrical Conductivity, and Piezoelectric Properties of Bismuth Titanate, J. Am. Ceram. Soc. 79 (1996) 3124–3128.
[14] H.S. Shulman, D. Damjanovic, N. Setter, Niobium doping and dielectric anomalies in bismuth titanate, J. Am. Ceram. Soc. 83 (2000) 528–532.
[15] X. Xie, T. Wang, Z. Zhou, G. Cheng, R. Liang, X. Dong, Enhanced piezoelectric properties and temperature stability of $Bi_4Ti_3O_{12}$-based Aurivillius ceramics via W/Nb substitution, J. Eur. Ceram. Soc. 39 (2019) 957–962.
[16] Z.Y. Shen, Z. Zhang, C. Qin, W. Luo, F. Song, Z. Wang, Y. Li, Effect of $(K_{0.5}Ce_{0.5})$ complex on the electrical properties of lead-free $Bi_4Ti_{2.86}W_{0.14}O_{12}$ high-temperature piezoelectric ceramics, J. Mater. Res. 36 (2021) 1134–1141.
[17] Y. Wu, J. Chen, J. Yuan, J. Xing, Z. Tan, L. Jiang, Q. Chen, J. Zhu, Structure refinements and the influences of *A*-site vacancies on properties of $Na_{0.5}Bi_{2.5}Nb_2O_9$-based high temperature piezoceramics, J. Appl. Phys. 120 (2016) 194103.
[18] Z. Zhou, R. Liang, Y. Li, X. Dong, Enhanced Electrical Resistivity of Al_2O_3 Addition Modified $Na_{0.5}Bi_{2.5}Nb_2O_9$ High-Temperature Piezoceramics, J. Am. Ceram. Soc. 98 (2015) 3925–3929.
[19] R. Aoyagi, H. Takeda, S. Okamura, T. Shiosaki, Synthesis and electrical properties of sodium bismuth niobate $Na_{0.5}Bi_{2.5}Nb_2O_9$, Mater. Res. Bull. 38 (2003) 25–32.
[20] C.-M. Wang, J.-F. Wang, High performance Aurivillius phase sodium-potassium bismuth titanate lead-free piezoelectric ceramics with lithium and cerium modification, Appl. Phys. Lett. 89 (2006) 202905.
[21] C.-M. Wang, J.-F. Wang, Aurivillius phase potassium bismuth titanate: $K_{0.5}Bi_{4.5}Ti_4O_{15}$, J. Am. Ceram. Soc. 91 (2008) 918–923.
[22] Q. Wang, E.-M. Liang, C.-M. Wang, High-performance bismuth titanate-ferrite ($Bi_5Ti_3FeO_{15}$) for high-temperature piezoelectric applications, J. Am. Ceram. Soc. 107 (2024) 4811–4823.
[23] Q. Wang, E.-M. Liang, C.-M. Wang, High performance bismuth titanate-ferrite piezoelectric ceramics for high-temperature applications, J. Eur. Ceram. Soc. 44 (2024) 5080–5087.
[24] Q. Wang, Z.-P. Cao, C.-M. Wang, Q.-W. Fu, D.-F. Yin, H.-H. Tian, Thermal stabilities of electromechanical properties in cobalt-modified strontium bismuth titanate ($SrBi_4Ti_4O_{15}$), J. Alloy. Compd. 674 (2016) 37–43.
[25] A.Z. Simões, C. Quinelato, A. Ries, B.D. Stojanovic, E. Longo, J.A. Varela, Preparation of lanthanum doped $Bi_4Ti_3O_{12}$ ceramics by the polymeric precursor method, Mater. Chem. Phys. 98 (2006) 481–485.
[26] Č. Jovalekić, M. Pavlović, P. Osmokrović, L. Atanasoska, X-ray photoelectron spectroscopy study of $Bi_4Ti_3O_{12}$ ferroelectric ceramics, Appl. Phys. Lett. 72 (1998) 1051–1053.
[27] K. Tang, W. Bai, J. Liu, J. Yang, Y. Zhang, C.-g Duan, X. Tang, J. Chu, The effect of Mn doping contents on the structural, dielectric and magnetic properties of multiferroic $Bi_5Ti_3FeO_{15}$ Aurivillius ceramics, Ceram. Int. 41 (2015) S185–S190.
[28] R. Pinho, A. Tkach, S. Zlotnik, M.E. Costa, J. Noudem, I.M. Reaney, P.M. Vilarinho, Spark plasma texturing: a strategy to enhance the electro-mechanical properties of lead-free potassium sodium niobate ceramics, Appl. Mater. Today 19 (2020) 100566.
[29] A.I. Spitsin, A.A. Bush, K.E. Kamentsev, Piezoelectric and dielectric properties of Bi_3TiNbO_9 prepared by hot pressing from powders activated using the serial dilution method, Sci. Rep. 10 (2020) 22198.
[30] Q. Wang, C.-M. Wang, Enhanced piezoelectric properties of Mn-modified $Bi_5Ti_3FeO_{15}$ for high-temperature applications, J. Am. Ceram. Soc. 103 (2020) 2686–2693.
[31] C. Moure, M. Villegas, A. Caballero, L. Lascano, P. Duran, J. Fernandez, Electrical behaviour of $Bi_5FeTi_3O_{15}$ and of their solid solution with $CaBi_4Ti_4O_{15}$, Boletin De. La Soc. Esp. De. Ceram. Y Vidr. 38 (1999) 577–581.
[32] Z. Huang, G.-S. Wang, Y.-C. Li, R.-H. Liang, F. Cao, X.-L. Dong, Electrical properties of (Na,Ce) doped $Bi_5Ti_3FeO_{15}$ ceramics, Phys. Status Solidi (a) 208 (2011) 1047–1051.
[33] A. Snedden, C.H. Hervoches, P. Lightfoot, Ferroelectric phase transitions in $SrBi_2Nb_2O_9$ and $Bi_5Ti_3FeO_{15}$: A powder neutron diffraction study, Phys. Rev. B 67 (2003) 092102.
[34] R.D. Shannon, Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides, Acta Cryst. A 32 (1976) 751–767.
[35] C.-M. Wang, J.-F. Wang, Z.-G. Gai, Enhancement of dielectric and piezoelectric properties of $M_{0.5}Bi_{4.5}Ti_4O_{15}$ (M=Na, K, Li) ceramics by Ce doping, Scr. Mater. 57 (2007) 789–792.
[36] M. Bian, Y. Ye, A. Cui, K. Jiang, W. Bai, H. Dong, B. Chen, Z. Hu, J. Chu, Phase transition of $Bi_5Ti_3FeO_{15}$ ceramics discovered by Raman spectroscopy and in situ synchrotron XRD under stress field, Appl. Phys. Lett. 117 (2020) 022901.
[37] J. Bobić, N. Ilić, V. Veerapandiyan, M.V. Petrović, M. Deluca, A. Dzunuzović, J. Vukmirović, K. Ning, K. Reichmann, S. Tidrow, Tailoring the ferroelectric and magnetic properties of $Bi_5Ti_3FeO_{15}$ ceramics by doping with Co and Y, Solid State Sci. 123 (2022) 106802.
[38] X. Mao, H. Sun, W. Wang, Y. Lu, X. Chen, Effects of Co-substitutes on multiferroic properties of $Bi_5FeTi_3O_{15}$ ceramics, Solid State Commun. 152 (2012) 483–487.
[39] V. Koval, Y. Shi, I. Skorvanek, G. Viola, R. Bures, K. Saksl, P. Roupcova, M. Zhang, C. Jia, H. Yan, Cobalt-induced structural modulation in multiferroic Aurivillius-phase oxides, J. Mater. Chem. C. 8 (2020) 8466–8483.
[40] X. Hu, W. Wang, X.-Y. Mao, X.-B. Chen, Magnetic and electric properties of Co-doped $Bi_5Ti_3FeO_{15}$ multiferroic ceramics, Acta Phys. Sin. 59 (2010) 8160–8166.
[41] Z. Yu, X. Meng, Z. Zheng, Y. Lu, H. Chen, C. Huang, H. Sun, K. Liang, Z. Ma, Y. Qi, T. Zhang, Room temperature multiferroic properties of rare-earth-substituted Aurivillius phase $Bi_5Ti_3Fe_{0.7}Co_{0.3}O_{15}$ ceramics, Mater. Res. Bull. 115 (2019) 235–241.
[42] P.P. Jiang, X.L. Zhang, P. Chang, Z.G. Hu, W. Bai, Y.W. Li, J.H. Chu, Spin-phonon interactions of multiferroic $Bi_4Ti_3O_{12}$-$BiFeO_3$ ceramics: Low-temperature Raman scattering and infrared reflectance spectra investigations, J. Appl. Phys. 115 (2014) 144101.
[43] T. Jardiel, A.C. Caballero, M. Villegas, Aurivillius ceramics: $Bi_4Ti_3O_{12}$-based piezoelectrics, J. Ceram. Soc. Jpn. 116 (2008) 511–518.
[44] C.-M. Wang, L. Zhao, Y. Liu, R.L. Withers, S. Zhang, Q. Wang, The temperature-dependent piezoelectric and electromechanical properties of cobalt-modified sodium bismuth titanate, Ceram. Int. 42 (2016) 4268–4273.
[45] X.-P. Jiang, X.-J. Wang, J.-X. Wen, C. Chen, N. Tu, X.-H. Li, Microstructure and electrical properties of Mn-modified bismuth-layer $Na_{0.25}K_{0.25}Bi_{2.5}Nb_2O_9$ Ceramics, J. Alloy. Compd. 544 (2012) 125–128.
[46] S. Guan, W. Shi, X. Wu, H. Xu, Y. Wu, R. Lang, Q. Chen, Optimized electromechanical performance in Aurivillius compounds via hot-pressing, J. Eur. Ceram. Soc. 44 (2024) 2087–2094.
[47] F. Huang, X. Lu, T. Xu, Y. Liu, W. Su, Y. Jin, Y. Kan, J. Zhu, Multiferroic properties of Co and Nd co-substituted $Bi_5Ti_3FeO_{15}$ thin films, Thin Solid Films 520 (2012) 6489–6492.
[48] J.D. Bobić, R.M. Katiliute, M. Ivanov, M.M. Vijatović Petrović, N.I. Ilić, A. S. Džunuzović, J. Banys, B.D. Stojanović, Dielectric, ferroelectric and magnetic properties of La doped $Bi_5Ti_3FeO_{15}$ ceramics, J. Mater. Sci. Mater. El. 27 (2016) 2448–2454.
[49] Z. Peng, D. Yan, Q. Chen, D. Xin, D. Liu, D. Xiao, J. Zhu, Crystal structure, dielectric and piezoelectric properties of Ta/W codoped Bi_3TiNbO_9 Aurivillius phase ceramics, Curr. Appl. Phys. 14 (2014) 1861–1866.

[50] F. Wang, X. Li, Q. Xu, H. Chen, J. Xi, F. Zhang, Z. Tan, Y. Cheng, S. Wang, J. Xing, Q. Chen, J. Zhu, Simultaneous enhancement of electrical and mechanical properties in $CaBi_2Nb_2O_9$-based ceramics, J. Eur. Ceram. Soc. 42 (2022) 4196–4211.

[51] Z. Yu, B. Yu, Y. Liu, P. Zhou, J. Jiang, K. Liang, Y. Lu, H. Sun, X. Chen, Z. Ma, T. Zhang, C. Huang, Y. Qi, Enhancement of multiferroic properties of Aurivillius $Bi_5Ti_3FeO_{15}$ ceramics by Co doping, Ceram. Int. 43 (2017) 14996–15001.

[52] D.Y. Suárez, I.M. Reaney, W.E. Lee, Relation between tolerance factor and T_c in Aurivillius compounds, J. Mater. Res. 16 (2001) 3139–3149.

[53] Y. Shimakawa, Y. Kubo, Y. Tauchi, T. Kamiyama, H. Asano, F. Izumi, Structural distortion and ferroelectric properties of $SrBi_2(Ta_{1-x}Nb_x)_2O_9$, Appl. Phys. Lett. 77 (2000) 2749–2751.

[54] Z. Li, M. Yang, J.-S. Park, S.-H. Wei, J.J. Berry, K. Zhu, Stabilizing Perovskite Structures by Tuning Tolerance Factor: Formation of Formamidinium and Cesium Lead Iodide Solid-State Alloys, Chem. Mater. 28 (2016) 284–292.

[55] X. Jiang, X. Jiang, C. Chen, N. Tu, Y. Chen, B. Zhang, Photoluminescence, structural, and electrical properties of erbium-doped $Na_{0.5}Bi_{4.5}Ti_4O_{15}$ ferroelectric ceramics, J. Am. Ceram. Soc. 99 (2016) 1332–1339.

[56] Z. Tan, S. Xie, L. Jiang, J. Xing, Y. Chen, J. Zhu, D. Xiao, Q. Wang, Oxygen octahedron tilting, electrical properties and mechanical behaviors in alkali niobate-based lead-free piezoelectric ceramics, J. Mater. 5 (2019) 372–384.

[57] M. Miyayama, Y. Noguchi, Polarization properties and oxygen-vacancy distribution of $SrBi_2Ta_2O_9$ ceramics modified by Ce and Pr, J. Eur. Ceram. Soc. 25 (2005) 2477–2482.

[58] Z.-P. Cao, C.-M. Wang, T.-L. Zhao, S.-L. Yu, H.-Z. Wu, Y.-M. Wang, Q. Wang, Y. Liang, Y.-N. Wei, Y. Zhang, Y. Liu, X.-S. Tang, Piezoelectric properties and thermal stabilities of strontium bismuth titanate ($SrBi_4Ti_4O_{15}$), Ceram. Int. 41 (2015) 13974–13982.

[59] C. Chen, K. Song, W. Bai, J. Yang, Y. Zhang, P. Xiang, M. Qin, X. Tang, J. Chu, Effect of Nb and more Fe ions co-doping on the microstructures, magnetic, and piezoelectric properties of Aurivillius $Bi_5Ti_3FeO_{15}$ phases, J. Appl. Phys. 120 (2016) 214104.

[60] R. Guo, L.E. Cross, S.E. Park, B. Noheda, D.E. Cox, G. Shirane, Origin of the High Piezoelectric Response in $PbZr_{1-x}Ti_xO_3$, Phys. Rev. Lett. 84 (2000) 5423–5426.

[61] Q. Tan, J. Li, D. Viehland, Role of lower valent substituent-oxygen vacancy complexes in polarization pinning in potassium-modified lead zirconate titanate, Appl. Phys. Lett. 75 (1999) 418–420.

[62] C. Long, H. Fan, M. Li, P. Ren, Y. Cai, A candidate for lead-free ultrahigh-temperature piezoelectrics: the excellent electro-mechanical properties of Aurivillius oxides, $Ca_{1-5x}Li_{2x}Nd_{3x}Bi_2Nb_{2-2x}Sc_xW_xO_{9-1.5x}$, CrystEngComm 15 (2013) 10212–10221.

[63] J. Yuan, R. Nie, Q. Chen, D. Xiao, J. Zhu, Structural distortion, piezoelectric properties, and electric resistivity of *A*-site substituted Bi_3TiNbO_9-based high-temperature piezoceramics, Mater. Res. Bull. 115 (2019) 70–79.

[64] C.-M. Wang, J.-F. Wang, S. Zhang, T.R. Shrout, Electromechanical properties of *A*-site (LiCe)-modified sodium bismuth titanate ($Na_{0.5}Bi_{4.5}Ti_4O_{15}$) piezoelectric ceramics at elevated temperature, J. Appl. Phys. 105 (2009) 094110.